Sec: SR STAR - A & APEX, ALL IITS | DT: 12/01/2026
Time: 3 Hrs | GTM – 16 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	A	A	B	C	C	B	C	C	D
11	12	13	14	15	16	17	18	19	20
D	A	C	D	C	A	D	D	B	A
21	22	23	24	25					
20	5	8	8	6					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	C	D	B	D	A	B	D	C	B
36	37	38	39	40	41	42	43	44	45
A	B	C	A	D	C	C	C	B	B
46	47	48	49	50					
75	5	5	3	6					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	A	B	C	A	A	D	A	B	C
61	62	63	64	65	66	67	68	69	70
B	A	C	A	A	C	A	B	A	D
71	72	73	74	75					
12	6	9	20	14					

Rankers Academy JEE
For More Join: @IITJEE_Advanc

HINTS
PHYSICS

1. As $\frac{1}{f}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$

$$\frac{1}{10}=(1.5-1)\left(\frac{1}{R}+\frac{1}{R}\right)=0.5\times\frac{2}{R}\Rightarrow R=10\ cm$$

$$\therefore \frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

Refraction from first surface,

$$\frac{1.5}{v_1}-\frac{1}{-20}=\frac{1.5-1}{+10}\Rightarrow v_1=\infty$$

For the second surface,

$$\frac{2}{v}-\frac{1.5}{\infty}=\frac{2-1.5}{-10}\Rightarrow v=-40cm$$

2. $\frac{K_1}{K_2}=\frac{m_2}{m_1}$

$$K_1=\frac{m_2}{m_1+m_2}\times mgl=\frac{6mgl}{7}$$

$$\frac{1}{2}mv_P^2=\frac{6mgl}{7}$$

$$V_P=\sqrt{\frac{12gl}{7}}$$

$$K_L=\frac{m_1}{m_1+m_2}=\frac{mgl}{7}$$

$$V_C=\sqrt{\frac{gl}{21}}$$

$$V_{rel}=V_P+V_C$$

For More Join: @IITJEE_Advanc

$$=\left(\sqrt{\frac{12}{7}}+\sqrt{\frac{1}{21}}\right)\sqrt{gl}$$

$$=\sqrt{\frac{3}{7}}\left(2+\frac{1}{3}\right)\sqrt{gl}$$

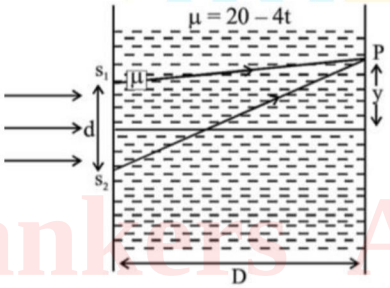
$$=\sqrt{\frac{7gl}{3}}$$

3. For central maxima, path diff $(\Delta X)=0$ for any point P on the screen.

X = thickness of glass slab.

$$\Delta X=\mu_m(S_2P)-\left[\mu_m(S_1P-X)+\mu X\right]$$

$$=\mu_m\left[S_2P-S_2P\right]-(\mu_m-\mu)X$$



$$=\mu_m\left(d.\frac{Y}{D}\right)-(\mu_m-\mu)X=0$$

Here,

$$y=\frac{D}{d}\left(\frac{\mu_m-\mu}{\mu_m}\right)X=\frac{DX}{d}\left(\frac{20-4t-5}{20-4t}\right)$$

$$=\frac{DX}{d}\left[\frac{15-4t}{20-4t}\right] \quad \text{.....(i)}$$

R.I. Of medium cannot b e less than 1 which become

At $t=\frac{19}{4}=4.75\,s$. Here after this time R.I of medium will not change. So position of centre maxima at time $t=5s$ will be same as time time $t=4.75s$.

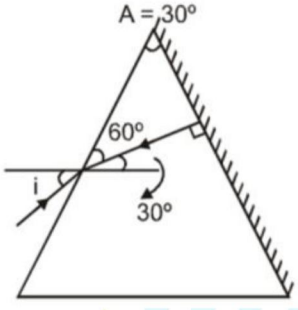
For More Join: @IITJEE_Advanc

$$\therefore y = \frac{DX}{d}\left(\frac{-4}{1}\right) = \frac{1}{2} \times 0.2 \times -4 = 0.4m$$

$$|y| = 40cm$$

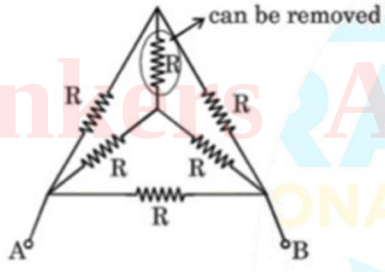
4. Conceptual

5.



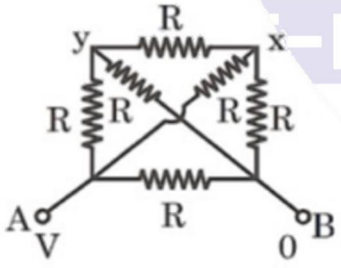
$$\frac{\sin i}{\sin 30°} = \sqrt{2} \Rightarrow \sin i = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}} \Rightarrow i = 45°$$

6. (Q) Equivalent diagram



$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{R}$$

$$\Rightarrow R_{eq} = \frac{R}{2}$$



(S) Assigning potential to each corner & solving

For More Join: @IITJEE_Advanc

$$R_{eq} = \frac{R}{2}$$

7. $z = \sqrt{(X_L - X_C)^2 + R^2} = \sqrt{25+25} = 5\sqrt{2}$

$$I_{rms} = \frac{10/\sqrt{2}}{5\sqrt{2}} = 1$$

$$V = 1 \times (X_L - X_C) = 1 \times 5 = 5 \text{ volt}$$

8. Component of velocity perpendicular to line AB must be equal

9. WD by $\vec{F} = \vec{C} \times \vec{V}$ will be zero

Since $WD = \int \vec{f}.d\vec{r} = \int \vec{f}.\vec{v}dt = \int (\vec{C} \times \vec{V}).\vec{V}dt = 0$

$$WD = \Delta K + \Delta U$$

$$0 = \left[\frac{1}{2}m\left(\frac{v_0}{2}\right)^2 - \frac{1}{2}mv_0^2\right] + qEd$$

10. $I = \frac{3}{2}mR^2$

Where,

$2\pi R = L$ and $m = \rho L$

$$= \frac{3}{8}\frac{\rho L^3}{\pi^2}$$

11. Conceptual

12. Acceleration of system $a = 5\ m/s^2$

In case of backward direction wave transmission:

Tension at point at r distance from C in the string $BC, T = 15 - 5r$

Wave velocity $v = \sqrt{\frac{T}{\mu}}$

$$\Rightarrow v = \sqrt{\frac{15-5r}{1}}$$

For More Join: @IITJEE_Advanc

$$\Rightarrow \frac{\partial r}{\partial t} = \sqrt{15 - 5r}$$

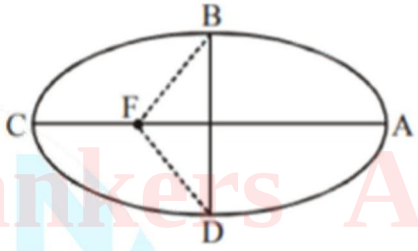
$$\Rightarrow \int_0^1 \frac{\partial r}{\sqrt{15 - 5r}} = \int_0^{t_1} \partial t$$

$$\Rightarrow \int_0^1 \frac{\partial r}{\sqrt{15 - 5r}} = \int_0^{t_1} \partial t$$

$$\Rightarrow t_1 = \frac{2}{5}\left(15^{1/2} - 10^{1/2}\right)$$

$$\Rightarrow t_1 = \frac{2}{\sqrt{5}}\left(\sqrt{3} - \sqrt{2}\right)\text{sec}$$

13.



$$\text{Focus} = -ae = -a\left(\frac{1}{2}\right)$$

$$b = a\sqrt{1 - e^2}$$

From conservation of angular momentum

$$mv_2 \frac{3a}{2} = mv \frac{\sqrt{3}}{2} a$$

$$v = \sqrt{3} v_2$$

Since areal velocity is constant during motion

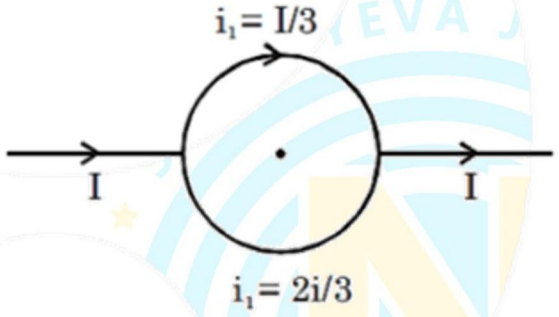
Time required will be proportional to area swept

$$\frac{T_{DAB}}{T_{BCD}} = \frac{\text{Area of FDAB}}{\text{Area of FBCD}}$$

For More Join: @IITJEE_Advanc

$$= \frac{\frac{\pi ab}{2} + \frac{1}{2}\sqrt{3}a\left(\frac{a}{2}\right)}{\frac{\pi ab}{2} - \frac{1}{2}\sqrt{3}(a)\left(\frac{a}{2}\right)}$$

14.



$$B_{centre} = \frac{\mu_0 I}{3\times 4r}\left(-\hat{k}\right) + \frac{\mu_0 2I}{3\times 4r}\left(\hat{k}\right)$$

$$-\frac{\mu_0 I}{12r}\hat{k} + \frac{2\mu_0 I}{12r}\hat{k}$$

$$B_{centre} = \frac{\mu_0 I}{12r}\hat{k}$$

15. $$b = \frac{ma}{k}\left[\sqrt{1+\frac{2kl}{ma}}\right]$$

Squaring both side and Apply dimension Rule

$$b^2 = \frac{m^2a^2}{k^2} \qquad b^2 = \frac{2lma}{K} \quad ....(2)$$

$$b = \frac{ma}{K} \quad ....(1) \qquad b^2 = 2lb$$

Putting (2) $b = 2l$ b=meter

$$b = L$$

16. $L_f = 0 \ \therefore \ L_i = 0$

$$mv_0R - \frac{mR^2}{2}\omega_0 = 0$$

For More Join: @IITJEE_Advanc

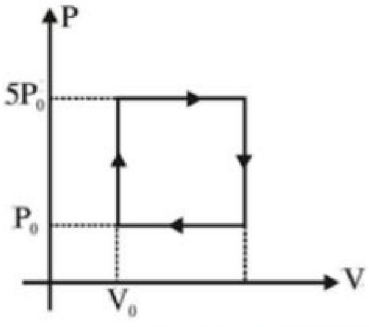
$$\therefore \frac{v_0}{r\omega_0} = \frac{1}{2}$$

17. The atom will absorb photon whose energy is equal to the energy gap between two energy levels of the atom.

18. $\lambda T = \text{constant}$

Area under curve $= \sigma A T^4$ (Radiated power is given not emissive power)

19.



$$\eta = \frac{\Delta w}{Q_s}$$

$$\eta = \frac{16P_0V_0}{\frac{5}{2}(4P_0V_0) + \frac{7}{2}(20P_0V_0)}$$

$$\eta = 0.2$$

20. Since soap film on A-side is broken

$\Rightarrow$ Soap film from B-side will pull the thread. Thus, thread will be concave towards A.

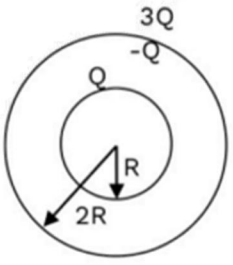
21. Potential energy of particle $= mgy$

$$\therefore F_x = \frac{-\partial U}{\partial x} = \frac{-\partial\left(20mgx^2\right)}{\partial x} = -40mgx$$

$$F \propto -x, \text{ thus } \omega^2 = \frac{40 \times 5 \times 10^{-5} \times 10}{5 \times 10^{-5}} \Rightarrow \omega = 20$$

For More Join: @IITJEE_Advanc

22.

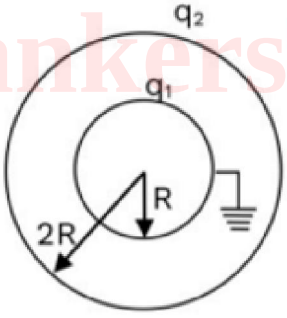


$$U_1 = \frac{Q^2}{2\left(\frac{4\pi\varepsilon_0 \times 2R^2}{R}\right)} + \frac{(3Q)^2}{2(4\pi\varepsilon_0 \times 2R)}$$

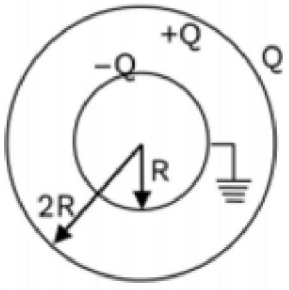
$$= \frac{Q^2}{16\pi\varepsilon_0 R} + \frac{9Q^2}{16\pi\varepsilon_0 R}$$

$$= \frac{10Q^2}{16\pi\varepsilon_0 R} = \frac{5Q^2}{8\pi\varepsilon_0 R}$$

After closing switch,



$$\frac{kq_1}{R} + \frac{kq_2}{2R} = 0 \; q_1 = \frac{-q_2}{2} = -\frac{2Q}{2} = -Q$$



For More Join: @IITJEE_Advanc

$$U_2 = \frac{Q^2}{16\pi\varepsilon_0 R} + \frac{Q^2}{16\pi\varepsilon_0 R} = \frac{Q^2}{8\pi\varepsilon_0 R}$$

$$\frac{U_1}{U_2} = \frac{5Q^2}{8\pi\varepsilon_0 R} \times \frac{8\pi\varepsilon_0 R}{Q^2} = 5$$

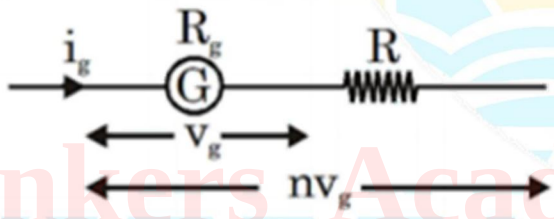
23. $v = k\sqrt{s} \Rightarrow v^2 = k^2 s \;\; \therefore a = \frac{k^2}{2}$

$$w = \frac{1}{2}mv^2$$

$$W = \frac{mk^4 t^2}{8}$$

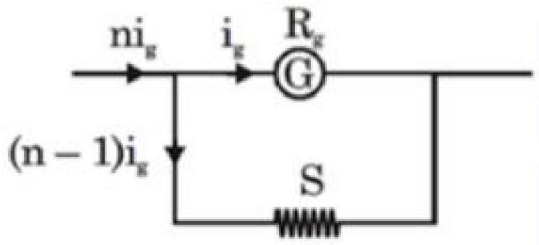
$$\therefore \beta = 8$$

24.



$i_g R_g = v_g$ and $i_g\left(R_g + R\right) = nV_g$

$\Rightarrow R = \left(n-1\right)R_g$ ....(i)



$i_g R_g = i_g\left(n-1\right)S$

$R_g = \left(n-1\right)S$ ....(ii)

$\frac{(i)}{(ii)} \rightarrow \frac{R}{R_g} = \frac{R_g}{S} \Rightarrow R_g = \sqrt{RS} = 8\Omega$

For More Join: @IITJEE_Advanc

25. If α emits, Z decreases by 2.

In β^+ decay, Z decreases by 1

In β^- decay, Z increases by 1

$7\alpha, 4\beta^- \ \& \ 2\beta^+$

$\Rightarrow Z$ decreases by $(7\times 2)+(2\times 1)-4\times 1$

$=14+2-4=12$

$\Rightarrow$ final $Z=72-12$

$=60=10N$

$\Rightarrow N=6$

CHEMISTRY

26. For mixing of ideal gases, entropy of the system increases. i.e., $\Delta S_{system} > 0$. The enthalpy change is zero, i.e., $\Delta H_{mixing} = 0$. Hence entropy of surroundings is also equal to zero.

27. $SF_4(sp^3d)$ / $SiF_4(sp^3)$ $S_2O_8^{2-}$ (sp^3) / $S_2O_3^{2-}$ (sp^3)

NO_3^- (sp^2) / NF_3 (sp^3) BrF_5 (sp^3d^2) / IF_7 (sp^3d^3)

28. i = 1 for solutions I, III and IV

$$\text{Molality m} = \frac{\text{Mole of solute}}{\text{Mass of solvent (in kg)}}$$

Molality order = ii < i< iii < iv

So F.P order = ii > i > iii > iv

B.P order = ii < i < iii < iv

29. Conceptual

30. Conceptual

31. Conceptual

32. The rate of $S_N1$reaction increases with increasing stability of carbocation.

33. Gibb's energy helps in predicting the best temperature and pressure conditions for industrial chemical processes to maximize the efficiency but not rate.

35. Conceptual

36. Conceptual

37. Aldol condensation of (B) gives the indicated product.

38. Conceptual

For More Join: @IITJEE_Advanc

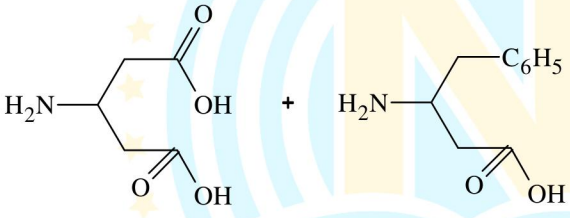
39. Rate law for the reaction, $-\frac{d[B_2]}{dt}=K[A][B_2]$ $\therefore K=\frac{1.8}{1.5\times1.5}$

$$-\frac{d[B_2]}{dt}=\frac{1}{2}\frac{d[AB]}{dt}=K[A][B_2]$$

40. Cannizzaro reaction is given by only aldehydes without containing α-hydrogen.

41. It has four functional groups, $-NH_2, -CO_2H, -\overset{O}{\overset{\|}{C}}-NH-$, and $-\overset{O}{\overset{\|}{C}}-O-CH_3$.

On hydrolysis it produces.

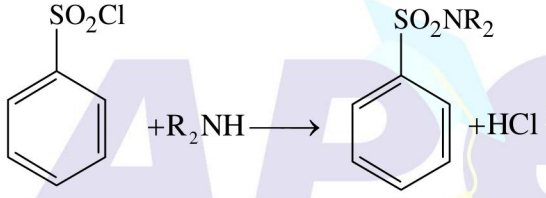


42. % of N in organic compound = $\frac{1.4\times N\times V}{w}=\frac{1.4\times10\times2\times1}{0.5}=56\%$

Also, milli equivalents of ammonia produced = milli equivalents of H_2SO_4 used

In case of 1M HCl, the % = $\frac{1.4\times10\times1}{0.5}=28\%$

43.



N, N dialkyl benzene sulphonamide is insoluble in KOH.

Conceptual

Most Pt 4–coordinated complexes are square planar

44. Conceptual

45. Conceptual

46. $POCl_3 + 3H_2O \longrightarrow H_3PO_4 + 3HCl$

50 ml 0.1 M $POCl_3$ = 5 m moles

1 mill mole product 3 m moles of $Ca(OH)_2$ required

For More Join: @IITJEE_Advanc

$\therefore$ 5 mill mole product requires 15 m moles $Ca(OH)_2$

$\therefore$ $V \times 2 = 15$

$\therefore$ $V = 7.5$ ml.

47. $\frac{hc}{\lambda} = 1 + hv_0 \ldots\ldots\ldots\ldots\ldots\ldots (1) \quad (1st\ light)$

$\frac{hc \times 3}{\lambda} = 4 + hv_0 \ldots\ldots\ldots\ldots\ldots (2) \quad (2nd\ light)$

From equation (1) and (2)

$hv_0 = 0.5\,eV$

48. i) All lanthanides exhibits +3 O.N. but few also exhibit +2 and +4 O.N.

ii) Correct

iii) Incorrect as all belongs to 6th period.

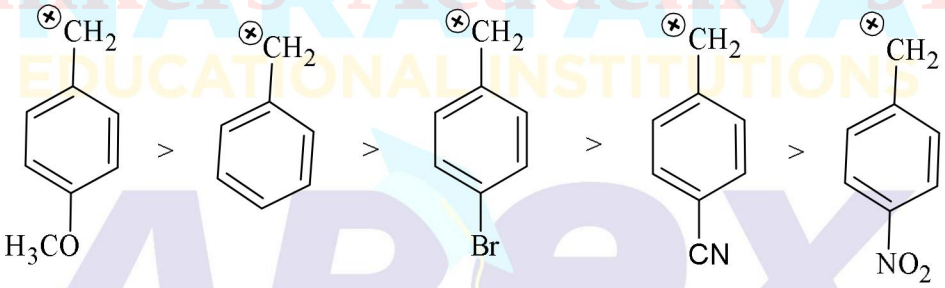
iv) True due to lanthanide contraction.

v) Incorrect as basicity decreases from $Ce(OH)_3$ to $Lu(OH)_3$ becomes of Fazan's rule.

vi) Correct

vii) All lanthanides have –ve S.R.P. hence all release hydrogen.

49. Substitution of —OH by Br atom takes place by carbocationic intermediate formation in rds. The stability of carbocation involved is as follows:



50. $i = 1.25$

original mole fraction = $\frac{1}{n} = \frac{1}{1+(n-1)}$

Now $\frac{1.25}{1.25+(n-1)} = \frac{1}{5}$

$6.25 = 1.25 + (n - 1)$

$n = 6$

For More Join: @IITJEE_Advanc

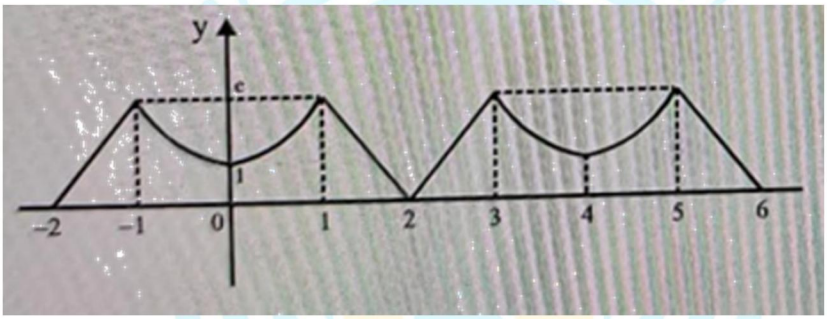
MATHEMATICS

51. $f'(x)+f(x)=12x^2+2$

52. For A $x-\frac{x^3}{6}<\sin x<x$ and for B use limit as a sum

$\Rightarrow A=\frac{1}{2},B=\frac{2}{\pi}$

53.



Graph of $f(x)$ is shown in two of its periodic interval.

$f(x)$ is non-derivable at $x=-9,-8,.....,0,1,.....9$

Range of $f(x):[0,e]$

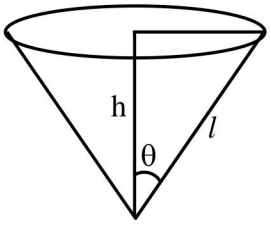
$f(x)$ is continuous at all points and non derivable at 19 points.

54. $f(x)=\begin{cases}x^2-x\sin x & x\geq 0\\ -x^2+x\sin x & x<0\end{cases}$

$f'(x)=\begin{cases}2x-x\cos x-\sin x & x\geq 0\\ -2x+x\cos x+\sin x & x<0\end{cases}$

$=\begin{cases}x(1-\cos x)+x-\sin x & x\geq 0\\ x(\cos x-1)-x+\sin x & x<0\end{cases}$

55.



For More Join: @IITJEE_Advanc

$$\tan\theta = \frac{5}{12} = \frac{r}{h} \Rightarrow r = \frac{5h}{12}$$

$$\frac{dv}{dt} = 0.1\ \text{m}^3/\text{min}$$

$$v = \frac{1}{3}\pi r^2 h = \frac{\pi 25h^3}{432}$$

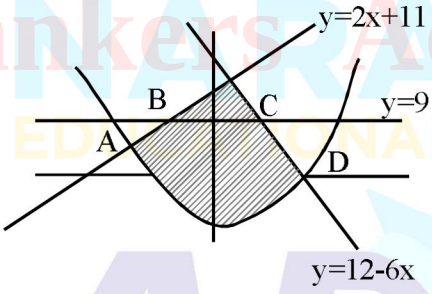
$$\Rightarrow \frac{dv}{dt} = \frac{25\pi h2}{144}\frac{dh}{dt} \Rightarrow 0.1 = \frac{25}{144} \times \pi \times 3 \times 3 \times \frac{dh}{dt}$$

$$\Rightarrow \frac{16}{250}\pi = \frac{dh}{dt}$$

Now $s = \pi rl = \pi r\sqrt{h^2 + r^2} = \frac{65\pi h^2}{144}$

$$\frac{ds}{dt} = \frac{65\pi}{144} \times 2h \times \frac{dh}{dt} = \frac{65\pi}{144} \times 2 \times 3 \times \frac{16}{250\pi} = \frac{13}{75}\text{m}^2/\text{min}$$

56.



$$x^2 - 4 = 2x + 11$$

$$x = -3, 5 \ \therefore A(-3,5)$$

$$2x + 11 = 9$$

$$x = 1 \ \therefore B(-1,9)$$

$$C\left(\frac{1}{2}, 9\right) \text{ and } D(2,0)$$

57. $f'(x) = 3x^2 - 3p^2x + 3p^2 - 3$

For More Join: @IITJEE_Advanc

$$=3\left(\left(x-p\right)^2-1\right)$$

$$=3\left(x-\left(p+1\right)\right)\left(x-\left(p-1\right)\right)$$

$$\Rightarrow 2>p-1>-4 \text{ and } -4<p+1<2$$

$$\Rightarrow -3<p<1$$

58. $$\frac{dy}{dx}=\frac{\left(x-2\right)+\left(y-2\right)}{\left(x-2\right)-\left(y-2\right)}$$

Put $x-2=h,\ y-2=k$

$$\frac{dk}{dh}=\frac{h+k}{h-k} \text{ put } k=vh$$

$$\Rightarrow v+h\frac{dv}{dh}=\frac{1+v}{1-v}$$

$$\Rightarrow \int\frac{1-v}{1+v^2}dv=\int\frac{dh}{h}\Rightarrow \tan^{-1}v=\frac{1}{2}\ln\left(1+v^2\right)+\ln h+c$$

$$\Rightarrow \tan^{-1}\left(\frac{y-2}{x-2}\right)=\frac{1}{2}\ln\left(1+\frac{\left(y-2\right)^2}{\left(x-2\right)^2}\right)+\ln\left(x-2\right)+c$$

$\because$ it passes through $\left(p+2,3\right)$

$$\tan^{-1}\left(\frac{1}{p}\right)=\frac{1}{2}\ln\left(1+\frac{1}{p^2}\right)+\ln p$$

$$\Rightarrow 2\tan^{-1}\left(\frac{1}{p}\right)=\ln\left(1+p^2\right)$$

59. $$T_r=\tan^{-1}\left(\frac{r.r!}{1+\left(r+1\right)!.r!}\right)$$

$$=\tan^{-1}\left(\left(r+1\right)!\right)-\tan^{-1}r!$$

60. $$f\left(e^4\right)=\int_1^{e^4}\frac{\log_{10}x}{1+x}dx$$

For More Join: @IITJEE_Advanc

Let $x=\frac{1}{v}\Rightarrow\frac{dx}{dv}=\frac{-1}{v^2}$

$$\Rightarrow f(t)=\int_1^{1/t}\frac{\log_{10}(1/v)}{1+\frac{1}{v}}-\frac{1}{v^2}dv=\int_1^{1/t}\frac{\log_{10}v}{v(v+1)}dv$$

$$\Rightarrow f(t)=\int_1^{1/t}\frac{\log_{10}v}{v(v+1)}\Rightarrow f\left(e^{-4}\right)=\int_1^{e^4}\frac{\log_{10}v}{v(v+1)}dv=\int_1^{e^4}\frac{\log_{10}x}{x(x+1)}dx \quad ....(2)$$

Add (1) and (2)

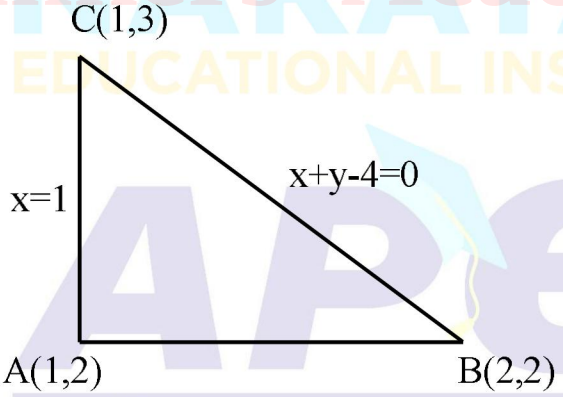
$$f\left(e^{-4}\right)+f\left(e^4\right)=\frac{1}{\ln 10}\int_1^{e^4}\frac{\ln x}{(1+x)\left(1+\frac{1}{x}\right)}dx$$

61. $(x+y-4)(xy-2x-y+2)=0$

$\Rightarrow(x+y-4)(x-1)(y-2)=0$

$\therefore$ sides of the triangle are $x+y-4=0,\ x=1, y=2$

$\therefore$ triangle is right angled at $(1,2)$.



Hypotenuse is $x+y=4,$ whose ends are $(1,3),(2,2)$.

$\therefore$ circumcentre is $\left(\frac{2+1}{2},\frac{2+3}{2}\right)$ i.e. $\left(\frac{3}{2},\frac{5}{2}\right)$

Circumradius $=\frac{1}{2}\sqrt{(1-2)^2+(3-2)^2}=\frac{1}{\sqrt{2}}$

For More Join: @IITJEE_Advanc

$$\Rightarrow x^2+y^2-3x-5y+\frac{9}{4}+\frac{25}{4}-\frac{1}{2}=0$$

$$\Rightarrow x^2+y^2-3x-5y+8=0$$

62. $E_1 \rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$$E_2=\frac{x^2}{A^2}\frac{y^2}{B^2}=1$$

Now as $x+y=3$ is a tangent

$$a^2+b^2=A^2+B^2=9$$

Now point P is

$$x^2+(2-x)^2=2$$

$$2x^2-4x+2=0$$

$$x=1$$

So P is $(1,2)$

Points Q and R are $\left(\frac{5}{3},\frac{4}{3}\right)$ and $\left(\frac{1}{3},\frac{8}{3}\right)$

Now $\left(\frac{5}{3},\frac{4}{3}\right)$ lies on E_1 so $\frac{25}{9a^2}+\frac{16}{9(9-a^2)}=1$

$$\Rightarrow 225-25a^2+16a^2=9a^2(9-a^2)$$

$$\Rightarrow 225-9a^2=9a^2(9-a^2)$$

$$\Rightarrow 25-a^2=a^2(9-a^2)$$

$$\Rightarrow a^4-10a^2+25=0 \Rightarrow a^2=5 \text{ so } b^2=4$$

$$e_1^2=\frac{1}{5}$$

$$\frac{1}{A^2}+\frac{64}{(9-A^2)}=9$$

For More Join: @IITJEE_Advanc

$9-A^2+64A^2=9A^2\left(9-A^2\right)$

$1+7A^2=A^2-9A^2-A^4\Rightarrow A^4-2A^2+1=0\Rightarrow A^2=1$ so $B^2=8$

$e_2^2=\frac{7}{8}$

63. The centre C of the circle is $(5,7)$

and radius $=\sqrt{5^2+7^2+51}=5\sqrt{5}$

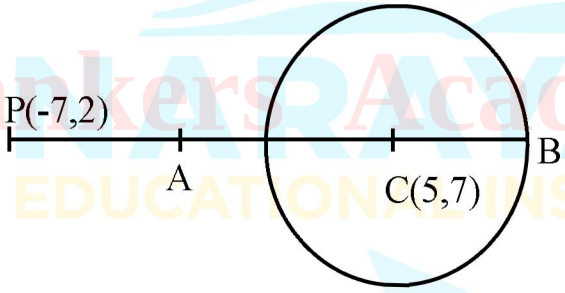
$PC=\sqrt{(5+7)^2+(7-2)^2}=13$

$\therefore PA=13-5\sqrt{5}$

And $PB=13+5\sqrt{5}$

$\therefore p=13+5\sqrt{5}$

$q=13-5\sqrt{5}$



G.M. of $p.q=\sqrt{pq}=\sqrt{169-125}=\sqrt{44}=2\sqrt{11}$

64. Equation of any line through the given pt. $(13,31)$ is $y-31=m(x-13)$

It meets the $x-$axis at $Q\left(13-\frac{31}{m},0\right)$ and y-axis at $S(0,31-13m)$

$\therefore$ the co-ordinates of R are (h,k), where $h=13-\frac{31}{m},k=31-3m$

$\therefore (h-13)(k-31)=31\times 13$

$\Rightarrow hk=31h+13k\Rightarrow \frac{13}{h}+\frac{31}{k}=1$

For More Join: @IITJEE_Advanc

Hence R satisfies $\frac{13}{x}+\frac{31}{y}=1$

65. Let the number of marble be 2n (where n is large)

Required probability = $\lim\limits_{n\to\infty}\frac{n\times {}^{n}C_4}{{}^{2n}C_5}\times\frac{{}^{n}C_3\times {}^{n}C_2}{{}^{2n}C_5}$

$$\lim_{n\to\infty}\frac{n\times n(n-1)(n-2)(n-3)}{4!}\times\frac{n(n-1)(n-2)}{3!}\times\frac{n(n-1)}{2!}\times\frac{(5)^2((2n-5)!)^2}{(2n!)^2}$$

$$\lim_{n\to\infty}\frac{n^4(n-1)^3(n-2)^2(n-3)((2n-5)!)^2\times 5\times 5\times 4\times 3!}{3!2!(2n!)^2}$$

$$\lim_{n\to\infty}\frac{50.n^4(n-1)^3(n-2)^2(n-3)}{(2n(2n-1)(2n-2)(2n-3)(2n-4))^2}=\frac{50}{1024}=\frac{25}{512}$$

66. Since $|z-1|=1$,

$\therefore$ let $z-1=\cos\theta+i\sin\theta$

Then, $z-2=\cos\theta+i\sin\theta-1$

$$=-2\sin^2\frac{\theta}{w}+2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}=2i\sin\frac{\theta}{2}\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right) \quad ....(i)$$

and $z=1+\cos\theta+i\sin\theta$

$$=2\cos^2\frac{\theta}{2}+2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}=2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right) \quad ....(ii)$$

From (i) and (ii), we get $\frac{z-2}{z}=i\tan\frac{\theta}{2}=\hat{i}\tan(\arg z)$

$\left(\because \arg z=\frac{\theta}{2}\text{ from }(ii)\right)$

67. Since given system of equation has a non-trivial solution

$$\therefore\begin{vmatrix}p+a & b & c\\ a & q+b & c\\ a & b & r+c\end{vmatrix}=0$$

For More Join: @IITJEE_Advanc

Operate $R_1 - R_2$ and $R_2 - R_3$, we get $\begin{vmatrix} p & -q & 0 \\ 0 & q & -r \\ a & b & r+c \end{vmatrix} = 0$

$$\Rightarrow p\begin{vmatrix} q & -r \\ b & r+c \end{vmatrix} + a\begin{vmatrix} -q & 0 \\ q & -r \end{vmatrix} = 0$$

$$\Rightarrow p(qr + qc + br) + a(qr) = 0$$

$$\Rightarrow 1 + \frac{c}{r} + \frac{b}{q} + \frac{a}{p} = 0 \qquad \Rightarrow \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = -1$$

68. Putting $x = -\alpha, x = -\beta, x = -\gamma$ in the equation

$f(-\alpha) = 1 + \cos(\beta - \alpha) + \cos(\gamma - \alpha) = 0$ .....(1)

$f(-\beta) = \cos(\alpha - \beta) + 1 + \cos(\gamma - \beta) = 0$ ....(2)

$f(-\gamma) = \cos(\alpha - \gamma) + \cos(\beta - \gamma) + 1 = 0$ .....(3)

From (1), (2), (3) we have

$\cos(\alpha - \gamma) = \cos(\alpha - \beta) = \cos(\gamma - \beta) = -1/2$

$\Rightarrow \gamma - \beta = 2\pi/3, \beta - \alpha = 2\pi/3, \gamma - \alpha = 4\pi/3$ is the only solution

69. Now $\vec{d}.(\vec{a} + \vec{b} + \vec{c}) = \vec{d}.\vec{a} + \vec{d}.\vec{b} + \vec{d}.\vec{c}$

$$= \mu(\vec{b} \times \vec{c}.\vec{a}) + \gamma(\vec{c} \times \vec{a}.\vec{b}) + \lambda(\vec{a} \times \vec{b}.\vec{c})$$

$$= \lambda + \mu + \gamma \qquad \left(\because [\vec{a}\ \vec{b}\ \vec{c}] = 1\right)$$

Hence $\lambda + \mu + v = \vec{d}.(\vec{a} + \vec{b} + \vec{c})$

70. The total no. of ways of predicting the result is 3^{10}. The no. of ways in which exactly K results are correct is ${}^{10}C_K.2^{10-K}$

$\therefore$ the no. of favourable ways is

$${}^{10}C_5.2^5 + {}^{10}C_6.2^4 + {}^{10}C_7.2^3 + {}^{10}C_8.2^2 + {}^{10}C_9.2 + {}^{10}C_{10} = 12585$$

Hence reqd. prob. $= \dfrac{12585}{3^{10}}$

71. Given eq. can be written as

For More Join: @IITJEE_Advanc

$$\left(\left(\sqrt{a^2-4}\right)\bar{i}+a\,\bar{j}+\left(\sqrt{a^2+4}\right)\bar{k}\right).\left(\left(TanA\right)\bar{i}+\left(TanB\right)\bar{j}+\left(Tan\,C\right)\bar{k}\right)=6a$$

$$\Rightarrow\left(\sqrt{a^2-4+a^2+a^2+4}\right)\left(\sqrt{Tan^2A+Tan^2B+Tan^2C}\right)\cos\theta=6a$$

$$\left\{\therefore \bar{a}.\bar{b}=|\bar{a}||\bar{b}|\cos\theta\right\}$$

$$\left(\sqrt{3}\,a\right).\sqrt{Tan^2A+Tan^2B+Tan^2C}=(6a)\sec\theta$$

$$Tan^2A+Tan^2B+Tan^2C=12\sec^2\theta$$

$$Sec^2\theta\geq 1\Rightarrow 12\sec^2\theta\geq 12$$

72. $4!.\left(2!.3+2\times 2+3\right)=312$

73. $\sigma^2=\dfrac{20\times 1+40\times 2^2}{60}=3$ (Because mean of both samples is the same)

74. $a=\left(4^{1/401}-1\right)$ and $b_n={}^nC_1+{}^nC_2.a+{}^nC_3.a^2+.....+{}^nC_n.a^{n-1}$

$$\therefore b_n=\frac{1}{a}\left[{}^nC_0+{}^nC_1a+{}^nC_2a^2+....{}^nC_na^n-1\right]$$

$$\therefore b_n=\frac{1}{a}\left[\left(1+a\right)^n-1\right]=\frac{4^{\frac{a}{401}}}{4^{\frac{1}{401}-1}}$$

$$\therefore b_{2006}-b_{2005}=\frac{4^{\frac{2006}{401}}-4^{\frac{2005}{401}}}{4^{1/401}-1}$$

$$=\frac{4^{\frac{2005}{401}}\left(4^{\frac{1}{401}}-1\right)}{\left(4^{\frac{1}{401}}-1\right)}$$

$$=4^5$$

75. $S_1=0$ for $n=1$

For More Join: @IITJEE_Advanc

$$S_n = \frac{n^2-1}{1-\frac{1}{n}} = n(n+1);\ n \geq 2$$

$$\sum_{n=1}^{\infty} \frac{S_n}{2^{n-1}} = S = \frac{2.3}{2} + \frac{3.4}{2^2} + \frac{4.5}{2^2} + \frac{5.6}{2^4} \ldots\ldots\infty$$

$$\Rightarrow \frac{S}{2} = \frac{2.3}{2^2} + \frac{3.4}{2^3} + \frac{4.5}{2^4} + \ldots..\infty$$

$$\therefore \frac{S}{2} = 3 + \frac{3}{2} + \frac{4}{2^2} + \frac{5}{2^3} + \frac{6}{2^4} \ldots\ldots\ldots\infty$$

$$\frac{S}{4} = \frac{3}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} \ldots\ldots\infty$$

$$\therefore \frac{S}{4} = 3 + 0 + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots\ldots\ldots\infty = \frac{7}{2}$$

$$\Rightarrow S = 14$$

Rankers Academy JEE
NARAYANA EDUCATIONAL INSTITUTIONS
APEX JEE-MAIN 2026

For More Join: @IITJEE_Advanc